The University of Nottingham
Malaysia Campus

SCHOOL OF COMPUTER SCIENCE

A LEVEL 1 MODULE, AUTUMN SEMESTER 2015-2016

SYSTEMS AND ARCHITECTURE

Time allowed ONE Hour

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*Candidates must **NOT** start writing their answers until told to do so*

Answer TWO questions out of THREE

Only silent, self-contained calculators with a single line display are permitted in this examination.

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Dictionaries are not allowed with one exception. Those whose first language is not English may use a standard translation dictionary to translate between that language and English provided that neither language is the subject of this examination. Subject specific translation dictionaries are not permitted.

No electronic devices capable of storing and retrieving text, including electronic dictionaries, may be used.

DO NOT turn examination paper over until instructed to do so

1. Networks (25 Marks)

(a) In the Aloha protocol, what happens if two stations attempt simultaneous transmission on the inbound frequency, and how is the problem handled?

[2 Marks]

(b) What happens to the operations of a token ring LAN if the intended recipient of a frame copies the frame into memory and then immediately crashes before it has a chance to set the A and C bits of the frame?

[4 Marks]

(c) Briefly explain the operations of the *CSMA/CA* protocol and explain why this protocol is needed for a wireless *LAN* access. [4 Marks]

(d) Suppose a network contains three Ethernet segments operating at 100 Mbps connected by two bridges and that each segment contains one computer. If two computers send a data to a third, what is the maximum and minimum data rate a given sender can achieve?

[2 Marks]

(e) With the help of diagrams, briefly explain the operations of a packet switching network using datagram approach. List and discuss the advantages and disadvantages of this approach. [7 Marks]

(f) Briefly describe the *static* and *dynamic* routing techniques. [4 Marks]

(g) Describe the purpose of associating a wireless computer with a specific base station. [2 Marks]

2. Digital Logic, Data Representation, and Memory (25 Marks)

(a) Using the truth table, verify the functions $(x + y')(y + z)$ and $(xy + xz + y'z)$ are equivalent.

[3 Marks]

(b) Given the Boolean function

$$F = (xy'z + x'y'z + xyz)$$

(i) List the truth table of the function.

(ii) Simplify the algebraic expression using Boolean algebra.

(iii) Draw the logic diagram using the simplified expression.

[6 Marks]

(c) Simplify the following function using four-variable Karnaugh map.

$F(A, B, C, D) = \Sigma\,(3, 7, 11, 13, 14, 15)$ [3 Marks]

(d) There are reasons for machine designers to want all instructions to have the same length. Why is this not a good idea in a stack machine? [2 Marks]

(e) Convert the decimal number $(12.125)_{10}$ into 32 bit binary number and represent the number by using a single precision IEEE floating-point format as shown in the following diagram.

Single precision format

1	8 bits	23 bits
Sign bit	Exponent in excess 127	Mantissa fraction

[4 Marks]

(f) Briefly describe the functions of *level 1 cache* and *level 2 cache* (i.e., L1 and L2 caches) memories in a computer. Describe why the memory-mapped I/O may fail to operate in a system with cache memory? [4 Marks]

(g) The 8-bits microprocessors and classic *CISC* computers like the Intel IA32 and 68K families have variable-length instructions. List one advantage and one disadvantage of computer architectures with instructions of different lengths.

[3 Marks]

3. Interrupts, Exception and I/O and Instruction Pipelining (25 Marks)

(a) A certain disk interface accepts requests to read a 32 KB block of data. It has a 32 KB buffer on board for its I/O interface, in which it stores the data as it comes off the drive. The interface is interrupt-driven and has DMA capability. Describe the likely sequence of events from the time the processor requests a block of data until the data have been transferred to the main memory. [6 Marks]

(b) Describe the terms, *memory-mapped I/O* and an *isolated I/O*. [4 Marks]

(c) When a device interrupt occurs, how does the processor determine which device has issued the interrupt?
[4 Marks]

(d) What, in the context of pipelined processors, is a *bubble* and why is it detrimental to the performance of a pipelined processor? [5 Marks]

(e) (i) Explain why branch operation reduce the efficiency of a pipelined architecture. [2 Marks]

(ii) Describe how branch prediction improves the performance of a *RISC* processor and minimizes the effect of branch instructions.
[4 Marks]